

Task 1: Student Marks Analysis

Input Data

```
students = [  
    ["Amit", 78, 85, 92],  
    ["Sara", 65, 72, 68],  
    ["John", 90, 88, 95],  
    ["Fatima", 45, 52, 48],  
    ["Ali", 80, 79, 81]  
]
```

Requirements

1. Calculate total marks.
 2. Calculate percentage.
 3. Find topper.
 4. Find students scoring below 50%.
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Task 2: Store Inventory

Input Data

```
products = [  
    ["Laptop", 25],  
    ["Mouse", 150],  
    ["Keyboard", 80],  
    ["Monitor", 18],  
    ["Printer", 5]  
]
```

Requirements

1. Find products with stock less than 20.
 2. Find total inventory count.
 3. Sort products by stock quantity.
 4. Create a suitable chart.
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Task 3: Employee Salaries

Input Data

```
employees = [  
    ["HR", 25000],  
    ["IT", 45000],  
    ["IT", 55000],  
    ["Sales", 30000],  
    ["Sales", 28000],  
    ["HR", 27000]  
]
```

Requirements

1. Calculate average salary department-wise.
2. Find highest paid employee.

3. Compare department salaries using a chart.
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Task 4: Missing Values Challenge

Input Data

```
data = {  
    "Name": ["A", "B", "C", "D", "E"],  
    "Age": [21, None, 23, None, 25],  
    "Marks": [78, 85, None, 90, 88]  
}
```

Requirements

1. Identify missing values.
 2. Handle them appropriately.
 3. Generate a clean dataset.
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Task 5: Monthly Sales

Input Data

```
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
```

```
sales = [25000, 30000, 28000, 40000, 42000, 38000]
```

Requirements

1. Calculate average sales.
 2. Find highest and lowest sales month.
 3. Plot sales trend.
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Task 6: Online Orders

Input Data

```
orders = [  
    [101, "Mobile", 2, 15000],  
    [102, "Laptop", 1, 55000],  
    [103, "Headphones", 3, 2000],  
    [104, "Tablet", 2, 18000]  
]
```

Requirements

1. Calculate bill amount for every order.
 2. Find order with highest value.
 3. Calculate total revenue.
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Task 7: Duplicate Records

Input Data

```
students = [  
    ["Amit", 22],  
    ["Sara", 21],  
    ["Amit", 22],  
    ["John", 23],  
    ["Sara", 21]  
]
```

Requirements

1. Detect duplicates.
 2. Remove duplicates.
 3. Display final dataset.
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Task 8: Temperature Monitoring

Input Data

```
temp = [32, 34, 35, 33, 38, 40, 42, 39, 37, 36]
```

Requirements

1. Average temperature.
 2. Hottest day.
 3. Coldest day.
 4. Visualize temperatures.
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Task 9: Exam Performance

Input Data

```
maths = [78, 85, 92, 65, 80]  
science = [82, 88, 90, 70, 75]
```

Requirements

1. Compare subject performance.
 2. Find correlation between subjects.
 3. Create scatter plot.
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